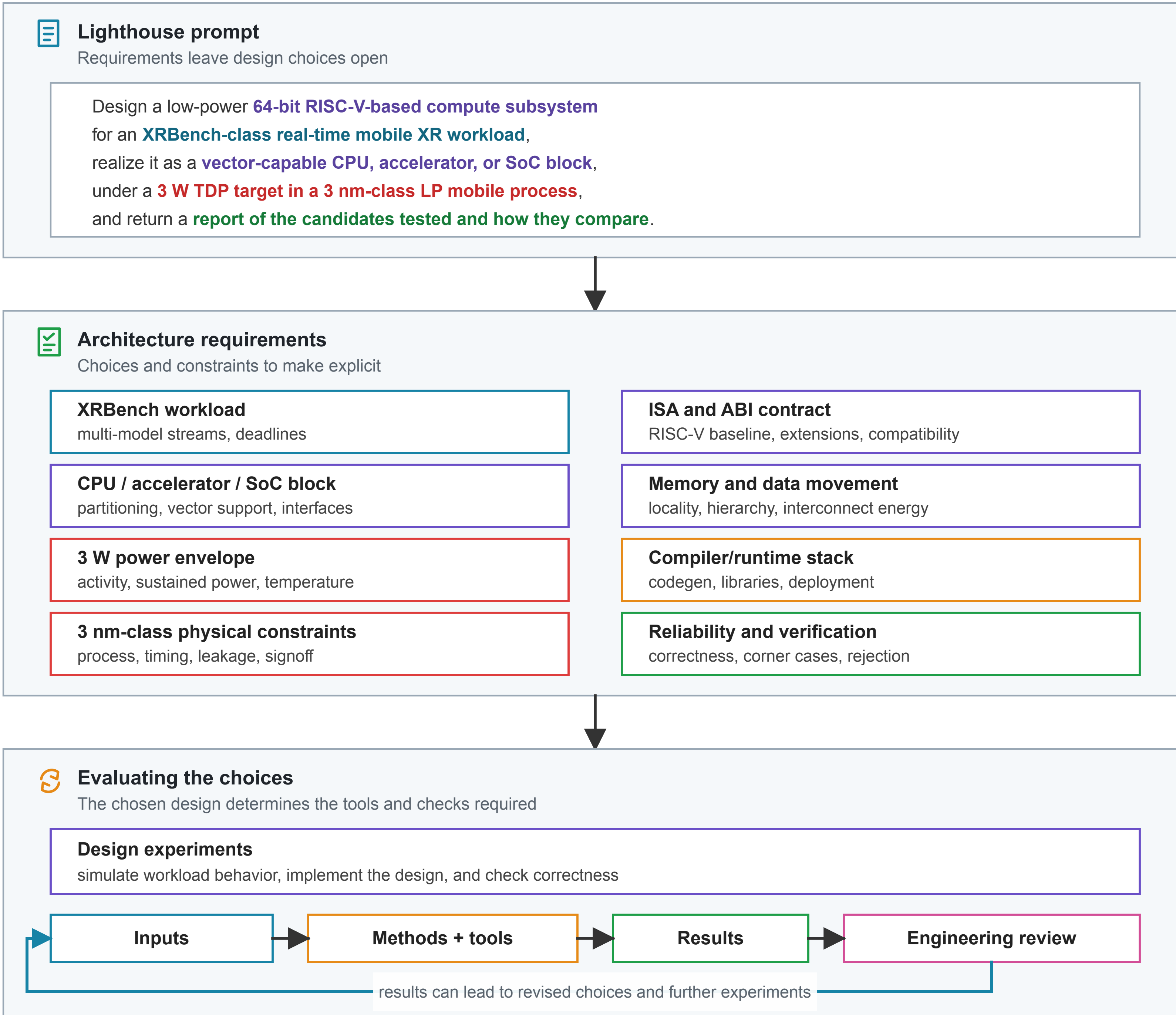

Lighthouse prompt
Requirements leave design choices open
Design a low-power 64-bit RISC-V-based compute subsystem
for an XRBench-class real-time mobile XR workload,
realize it as a vector-capable CPU, accelerator, or SoC block,
under a 3 W TDP target in a 3 nm-class LP mobile process,
and return a report of the candidates tested and how they compare.
Architecture requirements
Choices and constraints to make explicit
XRBench workload
multi-model streams, deadlines
ISA and ABI contract
RISC-V baseline, extensions, compatibility
CPU / accelerator / SoC block
partitioning, vector support, interfaces
Memory and data movement
locality, hierarchy, interconnect energy
3 W power envelope
activity, sustained power, temperature
Compiler/runtime stack
codegen, libraries, deployment
3 nm-class physical constraints
process, timing, leakage, signoff
Reliability and verification
correctness, corner cases, rejection
Evaluating the choices
The chosen design determines the tools and checks required
Design experiments
simulate workload behavior, implement the design, and check correctness
Inputs
Methods + tools
Results
Engineering review
results can lead to revised choices and further experiments